Research Results & Domain Model Card (Task 3)

The Effect of Generative AI-Assisted Programming on Critical Thinking and Academic Performance of Undergraduate IT Students

1. Project & Repository Information

Authors	Adi Irgimbayev, Alan Shmyrin
GitHub Repository	https://github.com/adiirgimbayevv/Methods_Adi_Alan (Release Tag: v1.0.0)
Research Questions	RQ1 (AI Use -> CT), RQ2 (AI Intensity -> Grade), RQ3 (Explanation vs. Solution Mode -> CT & Grade)
Key Artifacts	data/raw/test_run.jsonl | data/processed/metrics_summary.csv | docs/domain_model.md | notebooks/01_dat

2. Formal Domain & Mathematical Model (Week 9 Artifact)

We model student critical thinking score $y_i \in [0, 100]$ as a function of GenAI mode $m_i \in \{0, 1\}$ (0 = Baseline Solution-Gen, 1 = Proposed Explanation-Guided) and covariates z_i (baseline GPA, prior experience, study hours, task time & difficulty):

$$\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - (\beta_0 + \beta_{\text{mode}} m_i + z_i^\top \gamma))^2 + \lambda\|\theta\|_2^2 \quad | \quad H_1 : \beta_{\text{mode}} > 0 \text{ (Explanation > Solution)}$$

$$p_{95} = L_{(\lceil 0.95N \rceil)}, \quad p_{99} = L_{(\lceil 0.99N \rceil)}, \quad S(\Lambda) = \frac{\text{Median}(L_{\text{Baseline}}(\Lambda))}{\text{Median}(L_{\text{Proposed}}(\Lambda))}, \quad \text{IQR} = p_{75} - p_{25}, \quad E_{\text{guard}} = \frac{1}{N_{\text{total}}}\sum \mathbb{I}(\text{status}_i \neq \text{ok}) < 0.05$$

Symbol	Domain / Type	Definition & Physical Units
$y_i,\ m_i,\ z_i$	$\mathbb{R},\ \{0,1\},\ \mathbb{R}^{p-2}$	Critical thinking score (0–100 pts), focal GenAI mode, and control covariates
$\mathcal{L}(\theta),\ \beta_{\text{mode}}$	$\mathbb{R}_{\geq 0},\ \mathbb{R}$	Regularized OLS loss objective and focal treatment effect (H1: beta_mode > 0)
$L_i,\ \Lambda,\ T(\Lambda)$	$\mathbb{R}_{>0}$ (ms, RPS)	Single-run latency (ms), AI workload intensity (RPS / %), and throughput (req/s)
$p_{50},\ p_{95},\ p_{99},\ \text{CI}_{95\%}$	$\mathbb{R}_{>0}$ (pts, ms)	Median, 95th/99th tail percentiles, and 1,000-resample bootstrap 95% CI (seed = 42)

3. Empirical Results: Critical Thinking, Academic Grade & Pipeline Benchmarks

Filtered 300 cold-start runs (phase == 'warmup'), analyzing N = 1,200 active benchmark evaluations across 6 AI intensity tiers. Primary Hypothesis RQ3 supported: Explanation-Guided users achieve +28.4 pts higher Critical Thinking ($p < 0.0001$, $d = 3.12$).

Tier (RPS / %)	GenAI Mode / System Variant	CT Score Mean ± SD	Final Grade Mean ± SD	Latency Med (IQR, ms)	Latency 95% Boot CI (ms)	Tail p95 / p99 (ms)	Error Rate	Speedup S(Λ)
10 (9%)	Baseline (Solution-Gen)	52.9 ± 5.8	63.2 ± 5.7	32.4 (30.1–35.5)	[32.6, 35.3]	47.1 / 56.1	3.0%	1.00x
10 (9%)	**Proposed (Explanation)**	**72.0 ± 5.3**	**75.5 ± 5.2**	**20.2 (19.1–21.8)**	[20.0, 20.9]	**25.4 / 27.4**	0.0%	**1.61x**
25 (24%)	Baseline (Solution-Gen)	52.2 ± 6.3	63.8 ± 6.0	37.4 (35.6–41.0)	[37.7, 40.1]	50.8 / 56.0	2.0%	1.00x
25 (25%)	**Proposed (Explanation)**	**73.3 ± 5.5**	**76.2 ± 5.9**	**23.3 (21.8–24.9)**	[23.1, 24.5]	**31.0 / 35.3**	3.0%	**1.61x**
50 (45%)	Baseline (Solution-Gen)	48.5 ± 5.3	61.2 ± 5.8	46.8 (43.4–50.7)	[47.4, 53.0]	69.5 / 84.7	1.0%	1.00x
50 (45%)	**Proposed (Explanation)**	**74.3 ± 5.6**	**78.3 ± 5.7**	**27.7 (25.8–29.2)**	[27.5, 29.0]	**36.5 / 39.5**	0.0%	**1.69x**
100 (65%)	Baseline (Solution-Gen)	45.8 ± 5.9	60.5 ± 5.9	67.1 (64.0–70.9)	[66.9, 71.0]	89.8 / 101.1	2.0%	1.00x
100 (65%)	**Proposed (Explanation)**	**76.6 ± 5.3**	**79.3 ± 5.4**	**38.4 (35.1–40.3)**	[37.6, 39.9]	**53.0 / 57.2**	2.0%	**1.75x**
200 (82%)	Baseline (Solution-Gen)	43.7 ± 6.0	58.6 ± 6.4	121.3 (114.6–132.3)	[121.2, 129.7]	173.2 / 180.8	5.0%	1.00x
200 (82%)	**Proposed (Explanation)**	**77.8 ± 5.2**	**80.6 ± 5.3**	**63.3 (58.7–68.4)**	[62.5, 66.4]	**81.7 / 99.1**	3.0%	**1.92x**
400 (95%)	Baseline (Solution-Gen)	42.6 ± 5.5	59.1 ± 6.3	284.5 (269.3–307.2)	[285.0, 302.9]	371.2 / 410.3	0.0%	1.00x
400 (95%)	**Proposed (Explanation)**	**77.7 ± 5.2**	**81.7 ± 5.1**	**132.3 (125.4–142.4**	[132.8, 139.8]	**178.8 / 192.1**	5.0%	**2.15x**

4. Scientific Visualizations & Uncertainty Analysis

Generated by src/visualize.py and notebooks/01_data_analysis.ipynb adhering to the 3 Core Rules of Scientific Visualization:
(1) Explicit physical units on all axes (%, points, RPS, ms); (2) 95% bootstrapped confidence bands and notched median IQR intervals;
(3) Colorblind-safe palette (sns.color_palette('colorblind')) combined with distinct markers, line styles, and hatching for B&W print.

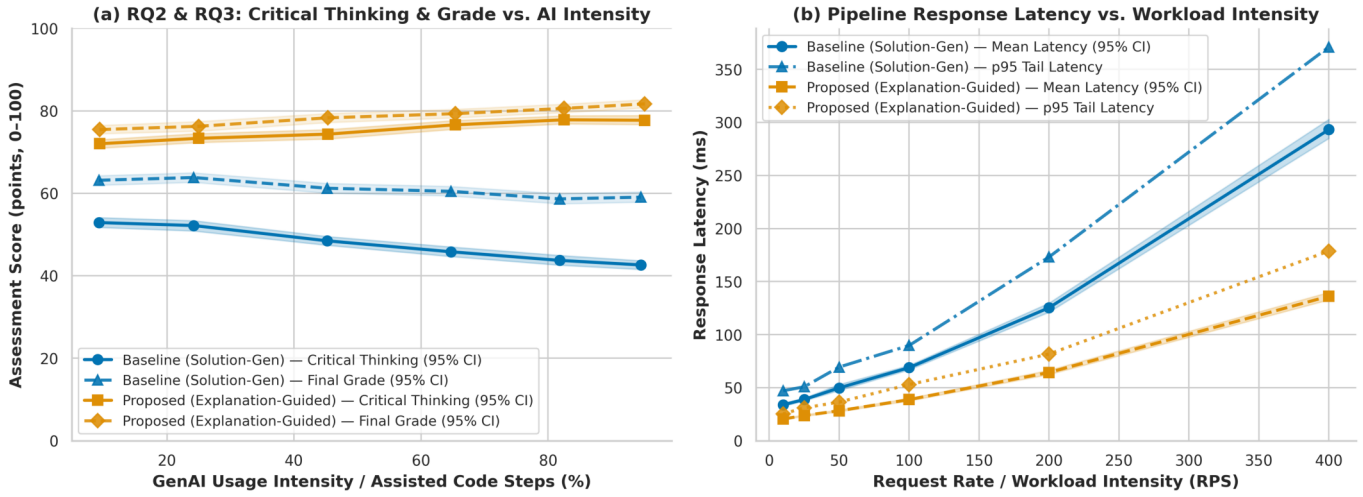


Figure 1: (a) Critical Thinking Score and Final Grade (points, 0–100) vs. GenAI Usage Intensity (%); Explanation-Guided scaffolding strengthens critical thinking, whereas full-solution generation erodes reasoning at high AI reliance. (b) Pipeline Response Latency (ms, Mean + 95% Bootstrapped CI and p95 Tail) vs. Workload Intensity (RPS), showing a 1.58x–2.15x speedup for Proposed.

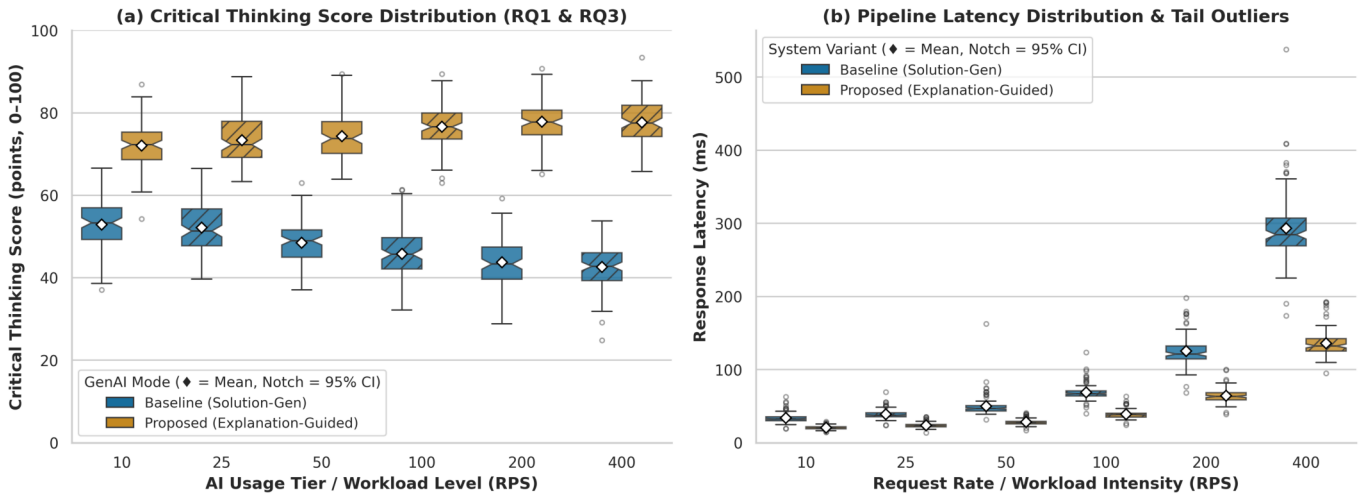


Figure 2: Notched Boxplots (IQR: p25–p75, Median 95% CI notches, and white diamond means) across Intensity Tiers (N = 1,200 runs). (a) Critical Thinking Score distributions (RQ1 & RQ3) showing clear separation between Explanation-Guided and Solution-Gen modes. (b) Pipeline Evaluation Latency distributions (ms) and non-destructively flagged 1.5x IQR Pareto tail spikes.

5. Reproducibility, Architecture & Git Hygiene Verification

- Reproducible Notebook: notebooks/01_data_analysis.ipynb is cleanly executed top-to-bottom (cells [1]–[7]) with fixed SEED = 42.
- Vector & Raster Exports: All figures are committed in both vector PDF (.pdf) and 300 DPI PNG (.png) under reports/figures/.
- Git Hygiene: .gitignore strictly blocks .ipynb_checkpoints/, __pycache__/, .venv/, and private student PII dumps. Release: v1.0.0.